Now, let your hands slowly approach again and then suddenly slam your hands back together. In this scenario, after the branes have drifted to their maximum distance and expanded, they are attracted back together, to bounce off each other again. That clap—the bounce—has destroyed everything on both branes, ended our universe, and created a new Big Bang. Both universes are back in the hot phase, filled with a plasma inferno, a chaotic state whose reborn space harbors little or no physical remnant of what it once may have held. Now separate your hands and do the whole cycle again. And again. And again. A braneworld[fn5] ekpyrotic universe is eternal, cataclysmic, cosmic applause.

ROUND AND ROUND AGAIN

Whether or not we really live on a braneworld, and whether or not there are other branes out there across some higher-dimensional bulk, are still open questions. The general idea of a cyclic universe holds some appeal, though, as it is one of a very few reasonable alternatives to inflation with any chance of duplicating its successes.[fn6] What shapes the ekpyrotic model and inflation will eventually take are yet to be determined—the newest ekpyrotic models don't require branes at all, while some versions of inflation now do. The big difference between ekpyrotic models and inflation is that where inflation solves a number of cosmological problems by introducing a period of rapid expansion in the very early universe, the ekpyrotic model does it through *slow contraction* just before the bounce. In the case of the braneworld model, this is during the phase when the branes are coming together. Like inflation, an ekpyrotic model might be compatible with the distribution of matter we see in the universe today, and can potentially explain why our cosmos appears to be so very uniform and flat (in the sense of not curving back on itself or having some other complicated large-scale geometry). The fact that everything is weirdly uniform makes sense if the branes are huge and parallel before the bounce —it means that the bang can happen everywhere in the same way at the same time, with only some slight quantum fluctuations adding the necessary blips to set up the higher-density regions that will grow up to become galaxies and clusters of galaxies and all of cosmic structure.

As with inflation, however, a lot of theoretical details are still being worked out. The biggest is the question of what exactly happens during the